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Examples

- A bird flying in the sky.
- Random motion of a gas molecule.
- Flying of a kite on a windy day.

2.3 ELEMENTARY CONCEPTS OF VECTOR ALGEBRA

In physics, some quantities possess only magnitude and some quantities possess both magnitude and direction. To understand these physical quantities, it is very important to know the properties of vectors and scalars.

Scalar

It is a property which can be described only by magnitude. In physics a number of quantities can be described by scalars.

Examples

Distance, mass, temperature, speed and energy

Vector

It is a quantity which is described by both magnitude and direction. Geometrically a vector is a directed line segment which is shown in Figure 2.10. In physics certain quantities can be described only by vectors.

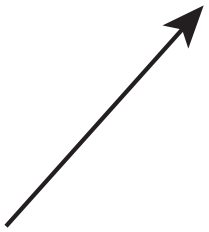


Figure 2.10 Geometrical representation of a vector

Examples

Force, velocity, displacement, position vector, acceleration, linear momentum and angular momentum

2.3.1 Magnitude of a Vector

The length of a vector is called magnitude of the vector. It is always a positive quantity. Sometimes the magnitude of a vector is also called 'norm' of the vector. For a vector $\vec{A}$, the magnitude or norm is denoted by $|\vec{A}|$ or simply 'A' (Figure 2.11).

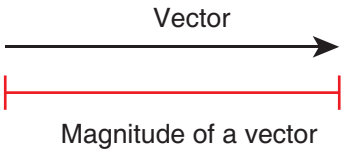


Figure 2.11 Magnitude of a vector

2.3.2 Different types of Vectors

1. **Equal vectors:** Two vectors $\vec{A}$ and $\vec{B}$ are said to be equal when they have equal magnitude and same direction and represent the same physical quantity (Figure 2.12.).

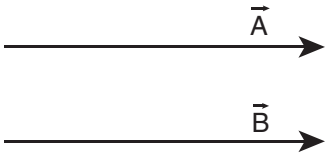


Figure 2.12 Geometrical representation of equal vectors

(a) **Collinear vectors**: Collinear vectors are those which act along the same line. The angle between them can be 0° or 180°.

(i) Parallel Vectors: If two vectors $\vec{A}$ and $\vec{B}$ act in the same direction along the same line or on parallel lines, then the angle between them is 0^0 (Figure 2.13).